

AI-POWERED MENTAL HEALTH STATUS ASSESSMENT

Deep Learning for Mental Health Monitoring from Digital Biomarkers

94.88%

TEST ACCURACY

1D Convolutional Neural Network using all 21 features

MODEL OUTPUT

Four predicted classes: Healthy, Mild_Stress, Moderate_Stress, and Severe_Stress.

The system returns a mental health status prediction with confidence probabilities.

01 | THE PROBLEM

- Manual mental health assessment is time-consuming and subjective.
- Affordable assessment tools are not always accessible.
- Existing systems may be proprietary or require specialized expertise.

02 | THE SOLUTION

- Automated mental health status prediction using a 1D CNN.
- Predicts four categories: Healthy, Mild_Stress, Moderate_Stress, and Severe_Stress.
- Uses 21 behavioral and physiological features.
- Provides fast results with confidence scores.

03 | HOW IT WORKS

- Input: 21 features — 3 scores plus 18 behavioral/physiological features.
- A 1D CNN analyzes patterns across the input features.
- Output: predicted mental health status and confidence probabilities.

04 | DATASET

- 4,000 samples across four mental health status classes.
- 21 features: 3 scores and 18 behavioral/physiological measures.
- Stratified 80/20 train/test split.
- Dropout regularization is used to help address overfitting and class imbalance.

05 | MODEL DESIGN

- 1D Convolutional Neural Network (CNN).
- Convolutional layers with BatchNormalization.
- Dropout regularization to help prevent overfitting.
- Adam optimizer with early stopping.

06 | EVALUATION & RESULTS

- Test accuracy: 94.88% using all 21 features.
- Evaluation includes classification reports.
- Confusion matrix and training history visualizations support model review.

07 | KEY FEATURES

- Single script with --train and --predict modes.
- Automatic evaluation metrics and reports.
- Command-line interface for training and prediction.
- Training history visualization.

08 | IMPACT

- Demonstrates deep learning for mental health assessment.
- Offers automated predictions to support healthcare decision-making.
- Serves as an educational tool for CNN applications.

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